

Multi-Factor Optimal Routing using Temporal Fusion Transformer Framework

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Abstract

Indian metropolitan cities face severe traffic congestion, a problem critically exacerbated by recurring exogenous shocks such as monsoon rains and festivals, leading to hazardous delays for emergency vehicles. Existing traffic management systems often fail in these high-variability conditions as they typically ignore external factors and provide single-objective deterministic routes. This paper proposes a multi-stage framework designed to address these limitations by integrating spatio-temporal traffic prediction with causal incident analysis and multi-objective routing. The core of our approach is a hybrid Temporal Fusion Transformer (TFT) and GNN model that generates uncertainty-aware traffic forecasts by explicitly incorporating weather and event data. We then employ a formal causal inference model to accurately quantify the real-world delay impact of road incidents. Finally, these outputs feed into a risk-aware routing engine that computes Pareto-optimal paths for emergency vehicles, balancing travel time and incident risk. The proposed framework is expected to deliver significant improvements in forecast accuracy and generate safer, faster, and healthier emergency routes.

Keywords: Traffic Forecasting, Emergency Routing, Causal Inference, Graph Neural Networks, Temporal Fusion Transformer, Multi-Objective Optimization.

1 Introduction

The rapid urbanization of Indian metropolitan centers such as Delhi, Mumbai, and Bengaluru has brought forth significant challenges in managing transportation infrastructure. Severe traffic congestion has become a daily reality, impeding economic productivity and, most critically, compromising public safety by delaying the response of emergency vehicles. This issue is magnified by the unique socio-environmental volatility of these regions. Unlike many Western cities where traffic patterns are more stable, Indian urban traffic is subject to extreme fluctuations driven by predictable shocks like major festivals and unpredictable ones like monsoon rains and frequent road incidents [2].

Existing traffic prediction and routing systems are often ill-equipped to handle this complex operational environment, exhibiting three fundamental limitations. First, they frequently fail to model crucial exogenous factors; the impact of heavy rainfall or a major public event is treated as noise rather than a primary predictive signal, leading to catastrophic forecast failures on the most critical days [1, 2]. Second, current systems typically model only the correlation between incidents and delays, failing to disentangle the true causal impact from confounding factors (e.g., an accident’s effect during peak vs. off-peak hours). This distinction is vital for accurate routing decisions, as a purely correlational model may misattribute delays and suggest suboptimal paths [8, 10]. Third, the common practice of providing a single, deterministic “fastest” route is inherently brittle. It ignores the intrinsic uncertainty in traffic prediction and fails to account for other critical objectives for an emergency response, such as minimizing the risk of encountering further incidents or limiting a patient’s exposure to harmful air pollution [2]. This context demands intelligent transportation systems (ITS) that are not merely accurate under average conditions but are resilient to these recurring, high-impact events [5].

The chronic nature of this congestion and its acute impact on emergency services can be viewed as a persistent, low-grade disaster. Therefore, the development of a solution requires a shift in perspective from mere prediction accuracy to overall system resilience. To address these challenges, this paper introduces a holistic framework that is:

- **Context-Aware:** By explicitly integrating weather, festival, and event data as first-class features in a state-of-the-art forecasting model.
- **Causally-Informed:** By moving beyond simple correlation to rigorously estimate the true causal impact of traffic disruptions using formal causal inference methodologies.
- **Risk-Aware and Multi-Objective:** By generating a set of optimal routes that balance speed with safety, predictability, and health-related factors, providing decision-makers with robust and actionable choices.

The remainder of this paper is organized to guide the reader through our research process. Section 2 establishes the context by reviewing the current state-of-the-art literature in traffic forecasting, causal inference, and emergency routing protocols. Following this foundation, Section 3 presents a comprehensive technical breakdown of our proposed five-stage framework, detailing the architecture and methodology. Section 4 defines our evaluation strategy, including the experimental setup and a discussion of the anticipated results. Finally, Section 5 concludes the study by summarizing the primary research contributions and outlining potential pathways for future work.

2 Literature Review

This research is built upon a foundation of prior work. The following review details the key papers and the specific aspects utilized from each paper.

Pearl [1] established the foundational theory of causal inference, emphasizing the critical distinction between statistical association and causation. This theoretical underpinning is essential for our framework, which aims to move beyond merely correlating traffic incidents with delays to identifying the actual causal impact of disruptions, thereby enabling more robust decision-making in emergency routing.

Lim et al. [2] introduced the Temporal Fusion Transformer (TFT) in the domain of deep learning for time series forecasting. This architecture is pivotal for our research as it provides a mechanism to integrate heterogeneous inputs—such as static metadata and time-varying exogenous factors like weather—into multi-horizon forecasts. The TFT’s ability to offer interpretable insights into temporal dynamics directly supports our goal of uncertainty-aware traffic prediction, particularly in the volatile context of Indian urban traffic.

Anandhalli et al. [3] proposed a deep learning method for pothole detection using Convolutional Neural Networks (CNNs) addressing specific infrastructure challenges. Their work highlights the granularity of risks present in Indian traffic conditions. In our framework, such localized road surface anomalies are treated as critical risk factors that must be accounted for when optimizing emergency vehicle routes to ensure both safety and speed.

Ismaeel et al. [4] utilized Deep Recurrent Neural Networks (RNNs) to classify traffic patterns in smart cities. Their success in capturing dynamic and sequential features of traffic flow validates the use of deep learning for categorizing urban mobility states. Our framework extends this by not just classifying patterns but using them as real-time inputs for risk-aware routing decisions.

Jiang et al. [5] provided a comprehensive review of Graph Neural Networks (GNNs) for traffic forecasting. They demonstrated that GNNs are uniquely suited to modeling the non-Euclidean topology of road networks. We leverage this insight by incorporating a GNN component into our prediction model to capture the spatial dependencies between connected road segments, ensuring network-wide awareness.

Huang et al. [6] applied the Temporal Fusion Transformer to flood risk prediction for substations. By successfully modeling multi-variable static and dynamic features, they demonstrated the TFT’s versatility in high-stakes environments. This supports our application of TFT to traffic scenarios where environmental factors like monsoon rains act as critical disruptors similar to flood risks.

Wu et al. [7] reviewed deep learning solutions for urban development challenges, emphasizing the integration of Bayesian regularization to handle uncertainty. Their focus on data privacy and safety in smart city applications informs our system design, ensuring that while we optimize for speed and efficiency, we also maintain robust and reliable model performance in complex urban settings.

Li et al. [8] proposed the Meteorological-Traffic Causal Inference Variational Auto-Encoder (MT-CIVAE) to estimate the causal impact of fine-grained weather

variations on traffic. This work is directly relevant to our causal modeling stage, providing a precedent for using deep learning to disentangle the specific effects of weather from other confounding variables in traffic dynamics.

Singh and Kumar [9] focused on enhancing emergency vehicle access in dense settlements using satellite imagery. Their deep learning approach to identifying road widths in informal settlements highlights the need for granular, localized data. We integrate this perspective by ensuring our routing engine considers physical constraints of the road network that are often overlooked in standard maps.

Zhong et al. [11] introduced a Large Language Model (LLM) system with "Chain-of-Thoughts" reasoning for traffic risk assessment. Their use of causal inference within an LLM to reduce hallucinations and improve decision reliability offers a novel approach to risk evaluation. We draw upon their methodology to enhance the interpretability of our risk scoring mechanism, ensuring that routing decisions are transparent.

Graham [10] provided a consolidated overview of causal inference specifically for transport research. This paper underscores the necessity of identifying causal effects in non-experimental settings, a common scenario in transport policy. We utilize the frameworks discussed by Graham to validate our causal modeling stage, ensuring our estimates of incident-induced delays are statistically rigorous.

Crowley and Ghosh [12] developed a Spatio-Temporal Graph Convolutional Network (STGCN) for urban traffic flow prediction. Their work confirms the efficacy of combining spatial and temporal modeling for capturing complex traffic dynamics. Our framework builds on this by fusing such spatio-temporal predictions with causal insights to optimize for multi-objective routing criteria.

Bisht et al. [13] utilized a Random Forest approach to correct biases in satellite precipitation estimates. This is critical for our data preprocessing stage, as accurate weather data is a key exogenous input for our traffic model. By adopting similar bias correction techniques, we ensure that the monsoon data feeding into our TFT model is reliable and representative of ground truth.

Finally, Fan et al. [14] addressed emergency logistics planning by optimizing multi-trip vehicle routing with limited trip duration. Their focus on equity and time windows in post-disaster scenarios aligns with our multi-objective routing goals. We adapt their constraints on trip duration and service times to ensure our emergency routes are not only fast but also operationally feasible under stress.

2.1 Research Gap

Despite the extensive work in traffic forecasting and deep learning, a significant research gap remains in the integration of these technologies for holistic emergency routing in highly volatile environments like Indian metropolises. While papers like [2, 5, 12] advance prediction accuracy using TFTs and GNNs, they largely treat traffic as a pure forecasting problem without explicitly modeling the *causal* mechanisms of disruption required for intervention. Conversely, causal inference studies like [1, 8, 10] provide the theoretical tools to understand "why"

delays happen but are rarely integrated into real-time, operational routing engines. Furthermore, existing emergency routing literature [9, 14] often focuses on static infrastructure or logistical constraints, neglecting the dynamic, multi-objective nature of risk that includes weather shocks [6, 13] and complex scene interpretation [11]. Our proposed framework fills this gap by creating a unified pipeline that not only forecasts traffic with uncertainty awareness but also causally quantifies incident impacts to generate routes that simultaneously optimize for time, safety, and environmental health.

3 The Proposed Spatio-Temporal Framework

3.1 Framework Architecture Overview

The proposed system is designed as a cohesive, multi-stage pipeline that transforms raw, heterogeneous data into actionable, multi-objective emergency route recommendations. The architecture, depicted in Figure 1, below, consists of five sequential stages: (1) Data Ingestion and Preprocessing, (2) Uncertainty-Aware Traffic Prediction, (3) Causal Modeling of Incident Impact, (4) Multi-Objective Routing, and (5) Deployment and Visualization. This layered approach ensures that each component builds upon the outputs of the previous one, creating a robust flow from data to decision. The following subsections provide more details about each of the stages.

3.1.1 Heterogeneous Data Ingestion and Preprocessing

The foundation of the framework is its ability to ingest and harmonize a diverse array of data sources. These include: vision-based traffic density data from Delhi, signal timing data from Bengaluru, Global Positioning System-based bus travel times from Delhi, meteorological data (rainfall, storms) from the India Meteorological Department (IMD), calendars of major festivals and public events, and the road network topology from OpenStreetMap (OSM).

The primary preprocessing step involves temporal alignment. All dynamic data streams (traffic, weather, etc.) are resampled and aligned to a common temporal resolution of 5-minute intervals. Following this, feature engineering is performed. Categorical variables, such as weather conditions ('light rain', 'heavy rain') and events ('Diwali', 'public holiday'), are one-hot encoded to create numerical features suitable for input into the deep learning model.

3.1.2 Uncertainty-Aware Traffic Prediction

The goal of this stage is to forecast future traffic conditions (speed, flow) across the road network while also quantifying the uncertainty of these predictions. We propose a hybrid Temporal Fusion Transformer (TFT) and Graph Neural Network (GNN) model to achieve this.

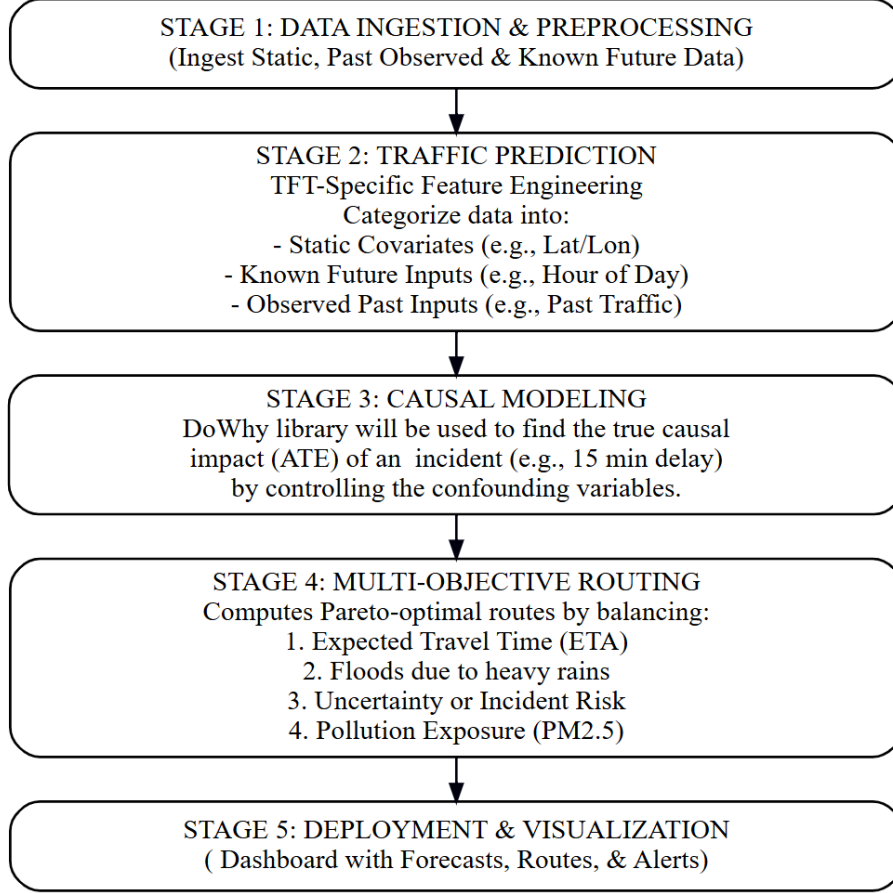


Figure 1: The Five Stage Approach to Implement TFT

First, the GNN component learns a rich spatial representation for each node (road segment) in the traffic graph. It operates on the OSM-derived graph, propagating information between adjacent road segments to capture spatial dependencies. Second, the TFT component takes these learned spatial embeddings as input, along with historical traffic data and the preprocessed exogenous features (weather, events). The TFT’s attention mechanism allows it to learn complex temporal patterns and identify the most influential variables for the prediction task.

Crucially, the model’s output is not a single point estimate. Instead, it is trained to produce a probabilistic forecast by predicting multiple quantiles (e.g., the 10th, 50th, and 90th percentiles) of the future traffic speed distribution. This is achieved by optimizing a quantile loss function. This probabilistic output directly provides the uncertainty bands (e.g., the difference between the 90th and 10th percentiles) that are a critical input for the risk-aware routing stage.

3.1.3 Causal Modeling of Incident Impact

This stage aims to estimate the Average Treatment Effect (ATE) of a traffic incident (e.g., accident, waterlogging) on travel time. This moves beyond correlation to provide a robust estimate of the delay directly caused by the incident. There are four steps in this stage. They are as follows.

1. **Model:** A causal graph based on domain knowledge, specifying the assumed causal relationships between variables has been defined. For instance, an incident (treatment) and time of day (confounder) both affect travel time (outcome).
2. **Identify:** Using the causal graph, DoWhy determines a valid statistical strategy to identify the causal effect, ensuring that the influence of confounding variables is properly controlled for.
3. **Estimate:** A statistical estimation method, such as propensity score matching or linear regression, is applied to the data to compute the ATE. The result is a quantified impact, such as "+15 minutes of delay caused by an accident on this road segment under current conditions".
4. **Refute:** DoWhy's built-in refutation methods have been used to test the robustness of our estimate against potential violations of our assumptions, such as the presence of an unobserved confounder.

3.1.4 Multi-Objective Emergency Routing

This stage synthesizes the outputs from the prediction and causal modeling stages to generate optimal emergency routes. The process begins by constructing a dynamic, weighted, directed graph $G = (V, E)$ from the OSM road network, where V is the set of intersections and E is the set of road segments.

The core innovation is the formulation of a **multi-objective edge weight** $\mathbf{W}_e$ for each edge $e \in E$. This weight represents a composite utility function that minimizes total path cost by combining four competing objectives into a **weighted summation**, as shown below:

$$\mathbf{W}_e = \alpha \cdot \mathbf{T}_e + \beta \cdot \mathbf{U}_e + \gamma \cdot \mathbf{R}_e + \delta \cdot \mathbf{P}_e \quad (1)$$

subject to $\alpha, \beta, \gamma, \delta \geq 0$ and $\alpha + \beta + \gamma + \delta = 1$

Where:

$\mathbf{W}_e$: The total multi-objective weight (cost) assigned to edge e .

$\mathbf{T}_e$: The Expected Travel Time on the edge, derived from the median (50th percentile) prediction of the TFT+GNN model.

$\mathbf{U}_e$: The Uncertainty (variability) of the travel time, calculated from the width of the predicted quantile range.

$\mathbf{R}_e$: The Incident Risk, which dynamically combines the probability of a new incident with the quantified causal delay impact.

$\mathbf{P}_e$: The PM2.5 Exposure for a patient traveling along the edge, derived from air quality data.

α, β, γ and δ are the weights trained on the historical datasets.

Since these objectives are often in conflict (e.g., the fastest route may be the most polluted or uncertain), there is no single "best" path. Therefore, we employ a multi-objective shortest path algorithm (such as a variant of A*) to compute the *Pareto-optimal set* of routes.

3.1.5 Interactive Visualization and Deployment

The final stage involves presenting the framework's outputs through an interactive dashboard for use in city traffic control centers. This dashboard, built with tools like Streamlit and Plotly, would provide a real-time operational view, including:

- Congestion heatmaps overlaid with uncertainty information.
- A map of active incidents, annotated with their estimated causal delay impacts.
- Recommended Pareto-optimal emergency routes, allowing operators to compare and select the most appropriate path for a given situation.

4 Evaluation Strategy and Expected Outcomes

The performance of the proposed framework has been validated through a comprehensive evaluation strategy targeting each of its core components. This includes quantitative metrics, ablation studies, and a clear set of anticipated performance gains.

4.1 Evaluation Metrics

To ensure a multifaceted assessment, we will employ a suite of metrics tailored to the specific task of each stage.

- **Prediction:** The accuracy of the traffic forecasts will be measured using standard regression metrics: Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE). The model's goodness of fit will be assessed with the R^2 score. To specifically evaluate the quality of the probabilistic forecasts, we will use the Continuous Ranked Probability Score (CRPS), which is a proper scoring rule that assesses both the accuracy and sharpness of the predicted distribution.

$$CRPS(F, x) = \int_{-\infty}^{\infty} (F(y) - \mathbf{1}(y \geq x))^2 dy \quad (2)$$

Where:

- $\mathbf{x}$ is the observed outcome (the actual realized value, such as measured traffic speed).
- $F(y)$ is the Cumulative Distribution Function (CDF) of the forecast, representing the predicted probability distribution.
- $\mathbf{1}(y \geq x)$ is the indicator function, which equals 1 if the condition is met (i.e., the forecast is less than the observation), and 0 otherwise.
- **Causal Modeling:** The output of the causal inference stage will be evaluated by the Average Treatment Effect (ATE) and its associated Confidence Intervals. This provides not only a point estimate of an incident’s impact but also a measure of its statistical certainty.

$$ATE = E[Y(1) - Y(0)] \quad (3)$$

Where:

- $E[.]$ denotes the Expected value (average).
- $Y(1)$ is the potential outcome with treatment (e.g., travel time with an incident).
- $Y(0)$ is the potential outcome without treatment (e.g., travel time without an incident).
- **Routing:** The quality of the generated routes will be evaluated against multiple criteria: Mean Estimated Time of Arrival (ETA) to assess speed, the 95th percentile ETA to measure worst-case reliability, the percentage of high-risk edges avoided to quantify safety, and a cumulative PM2.5 exposure index to measure health benefits.

Furthermore, a series of ablation studies will be conducted to isolate the contribution of each novel component of the framework. By systematically removing the weather/event features, the uncertainty component, or the causal impact analysis, we can quantitatively demonstrate the value added by each innovation.

4.2 Anticipated Performance Gains

Based on the novel architecture and integrated approach, we hypothesize a set of significant improvements over baseline systems. These expected outcomes, summarized in Table 1, represent the primary performance targets of this research. The integration of context-specific data is expected to yield more accurate forecasts, especially during critical high-congestion periods. The use of causal inference will provide more realistic inputs for routing, and the multi-objective, uncertainty-aware pathfinding is projected to produce emergency routes that are not only faster on average but also more reliable and safer.

Table 1: Projected Performance Improvements of Proposed Framework

Component	Key Benefit	Projected Improvement
Traffic Prediction	Integration of monsoon/event data improves forecast accuracy on critical days.	RMSE ↓ 15–20%
Incident Analysis	Causal inference provides realistic, quantified delay impacts per incident type.	ATE with CIs
Emergency Routing	Multi-objective, uncertainty-aware pathfinding reduces ETA variance and risk exposure.	95th percentile ETA; Avoidance of risky/polluted segments
System Utility	Interactive dashboard provides actionable insights for traffic operators.	Enhanced situational awareness

5 Conclusion

This paper presented a comprehensive framework for emergency vehicle routing in Indian metropolitan cities. Addressing limitations regarding exogenous shocks, correlational models, and single-objective routing, our solution offers a resilient, intelligent system. It integrates state-of-the-art machine learning, causal inference, and multi-objective optimization into a unified pipeline. Primary contributions include: the first India-specific prediction system integrating monsoon and festival data; the application of causal inference to quantify incident delays beyond correlation; and an uncertainty-aware routing engine balancing ETA, safety, and patient health for robust, Pareto-optimal routes. Future work could explore the extension of this framework to include real-time traffic control actions, such as dynamic traffic signal timing, or its application to other complex urban logistics problems beyond emergency response.

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